Práctica 2: Sistema cardiovascular

Departamento de Ingeniería Eléctrica y Electrónica, Ingeniería Biomédica

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Table of Contents

Información general....1
Datos de la simulación....1
Respuestas: Normotenso, Hipotenso e Hipertenso....2
Respuestas: Normotenso, Hipotenso y Controlador....2
Respuestas: Normotenso, Hipertenso y Controlador....3
Modelo de EDOs....3
Función: Respuestas de simulación....4

Información general



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Asignatura: **Modelado de Sistemas Fisiológicos**

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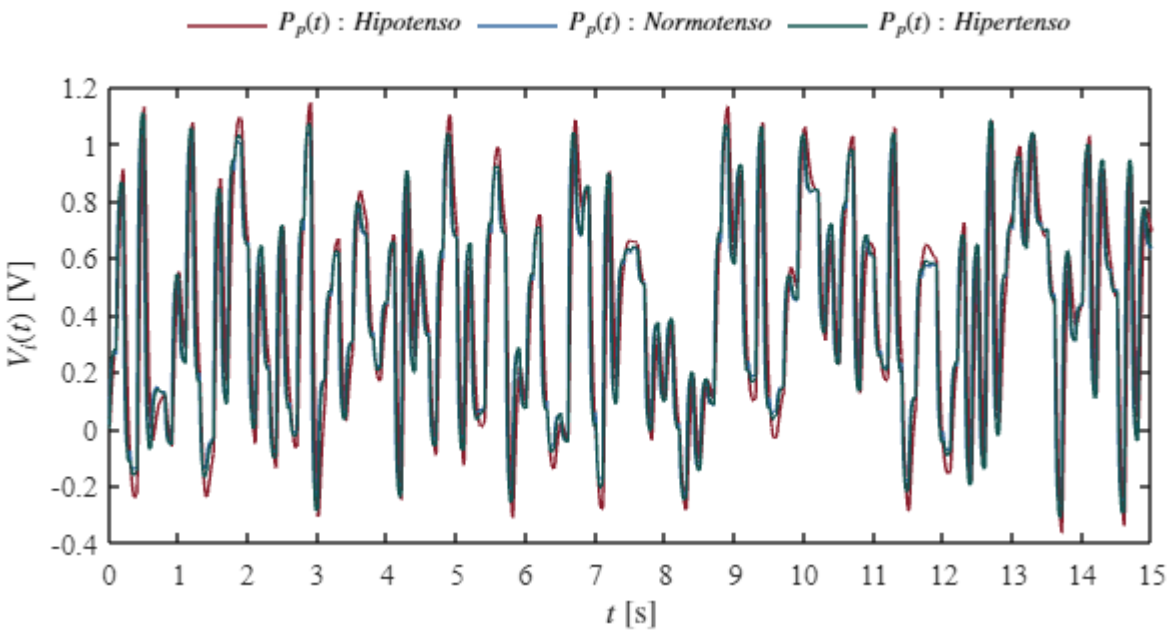
Datos de la simulación

```
clc; clear; close all; warning('off','all')
open_system('sistema')
parameters.StopTime = '15';
parameters.Solver = 'ode15s';
parameters.MaxStep = '1e-3';
```

```
x = sim('sistema',parameters);
t_data = (0:0.1:15)';
t = (0:0.001:15)';
Pa = interp1(t_data, x.Pa, t, 'previous');
writematrix(x.Pa,'signal.xlsx');
```

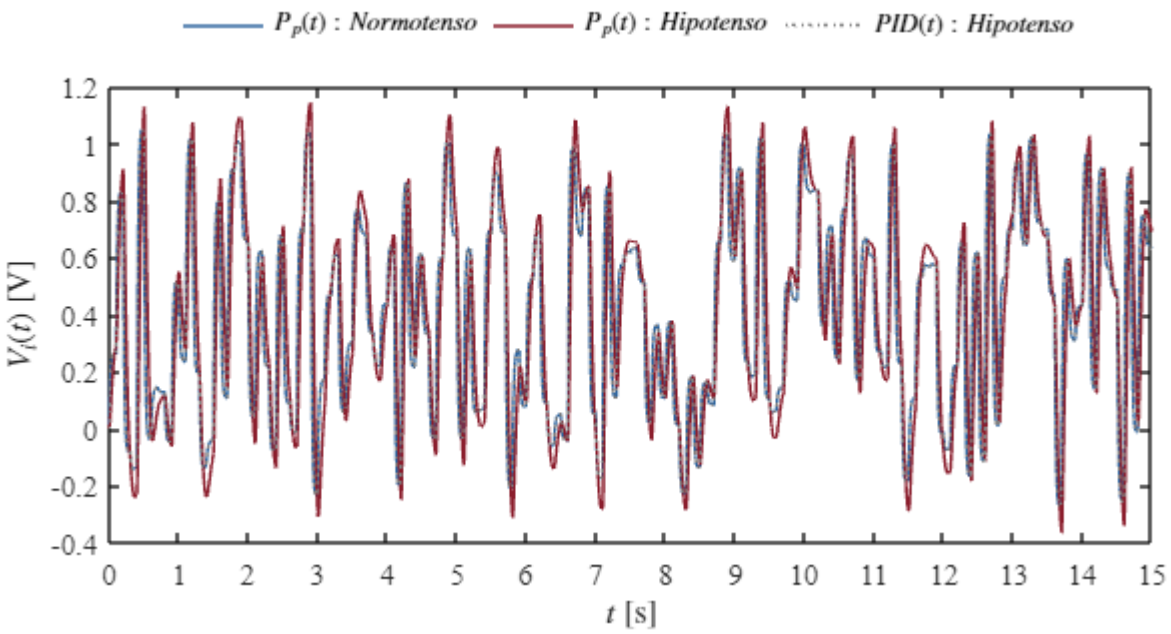
Respuestas: Normotenso, Hipotenso e Hipertenso

```
plotsignals(x.t,x.PpL,x.PpN,x.PpH,'individuos')
```



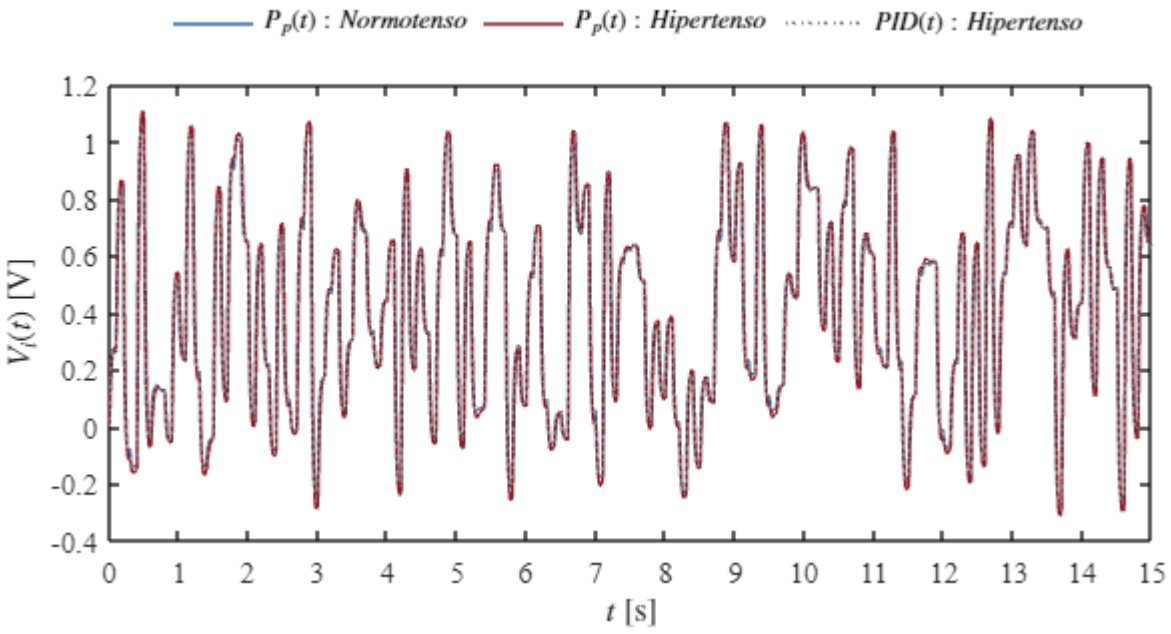
Respuestas: Normotenso, Hipotenso y Controlador

```
plotsignals(x.t,x.PpN,x.PpL,x.PIDL,'hipotenso')
```



Respuestas: Normotenso, Hipertenso y Controlador

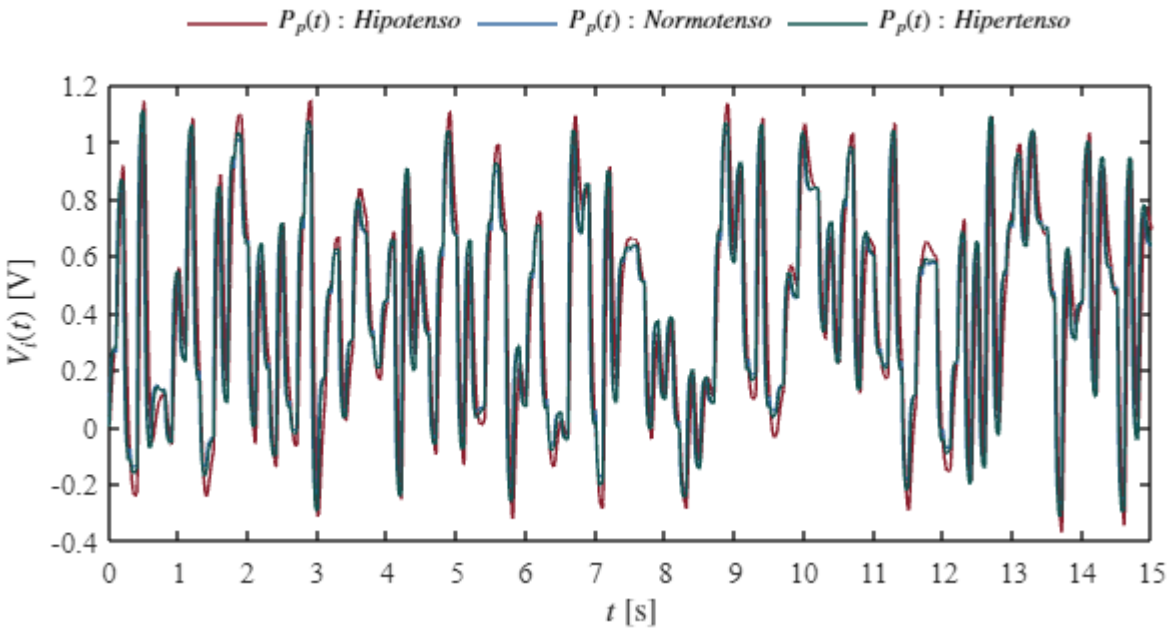
```
plotsignals(x.t,x.PpN,x.PpH,x.PIDH,'hipertenso')
```



Modelo de EDOs

```
par_hipo = [0.01,0.001,1.00,3.500]; %[Z,L,R,C]
par_normo = [0.01,0.001,1.00,1.500];
par_hiper = [0.01,0.001,1.00,1.000];
PpN = cardio(par_normo,Pa,t);
PpL = cardio(par_hipo,PpN,t);
```

```
PpH = cardio(par_hiper,PpN,t);
plotsignals(t,PpL,PpN,PpH,'odes')
```




```
function Pp = cardio(par,Pa,t)
    Z = par(1); L = par(2); R = par(3); C = par(4); dt = t(2);
    n = round(t(end)/dt);
    FL = zeros(length(t),1); FL(1) = 0;
    Pp = zeros(length(t),1); Pp(1) = 0;
    for i = 1 :n
        FL(i+1) = FL(i) + (-Pp(i)/L + Pa(i)/L)*dt;
        Pp(i+1) = Pp(i) + (FL(i)/C - (1/(Z*C) + 1/(R*C))*Pp(i) + Pa(i)/(Z*C))*dt;
    end
end
```

Función: Respuestas de simulación

```
function plotsignals(t,x1,x2,x3,caso)
    set(figure(),'color','w')
    set(gcf,'units','inches','position',[1,1,7,3.5])
    set(gca,'fontname','times new roman','fontsize',11)
    hold on; grid off; box on;
    clr1 = [139, 30, 45] / 255;
    clr2 = [51, 104, 160] / 255;
    clr3 = [18, 84, 79] / 255;
    clr4 = [0.5, 0.5, 0.5];

if caso == "individuos"
    plot(t,x1,'-','LineWidth',0.25,'Color',clr1) %Hipotenso
```

```
    plot(t,x2,'-','LineWidth',0.25,'Color',clr2) %Normotenso
    plot(t,x3,'-','LineWidth',0.25,'Color',clr3) %Hipertenso
    L = legend('$P_p(t):Hipotenso$','$P_p(t):Normotenso$','$P_p(t):Hipertenso$');
elseif caso == "hipotenso"
    plot(t,x1,'-','LineWidth',0.25,'Color',clr2) %Normotenso
    plot(t,x2,'-','LineWidth',0.25,'Color',clr1) %Hipotenso
    plot(t,x3,':','LineWidth',1,'Color',clr4) %Controlador del hipotenso
    L = legend('$P_p(t):Normotenso$','$P_p(t):Hipotenso$','$PID(t):Hipotenso$');
elseif caso == "hipertenso"
    plot(t,x1,'-','LineWidth',0.25,'Color',clr2) %Normotenso
    plot(t,x2,'-','LineWidth',0.25,'Color',clr1) %Hipertenso
    plot(t,x3,':','LineWidth',1,'Color',clr4) %Controlador del hipertenso
    L = legend('$P_p(t):Normotenso$','$P_p(t):Hipertenso$','$PID(t):Hipertenso$');
elseif caso == "odes"
    plot(t,x1,'-','LineWidth',0.25,'Color',clr1) %Hipotenso
    plot(t,x2,'-','LineWidth',0.25,'Color',clr2) %Normotenso
    plot(t,x3,'-','LineWidth',0.25,'Color',clr3) %Hipertenso
    L = legend('$P_p(t):Hipotenso$','$P_p(t):Normotenso$','$P_p(t):Hipertenso$');
end

set(L,'interpreter','latex','location','northoutside','orientation','horizontal','bo
x','off')

xlabel('$t$ [s]','interpreter','latex')
ylabel('$V_i(t)$ [V]','interpreter','latex')

xlim([0 15]); xticks(0:1:15)
ylim([-0.4 1.2]); yticks(-0.4:0.2:1.2)

exportgraphics(gcf,[caso,'_matlab.pdf'],'contenttype','vector')
end
```